B. Design of FIR filter : Windows Technique

Let us consider that the digital filter which is to be designed have the frequency response $H_d(\ )$ which is also called desired frequency response. Let the corresponding unit sample response that is desired to be $h_d[n]$. But, $H_d(e^{j\omega})$ is simply the Fourier Transform of $h_d[n]$

This means that,

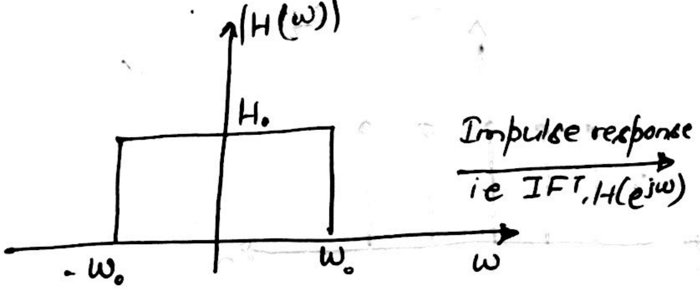
$$H_d(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h_d[n]\, e^{-j\omega n} \quad \text{——— (1)}$$

where,

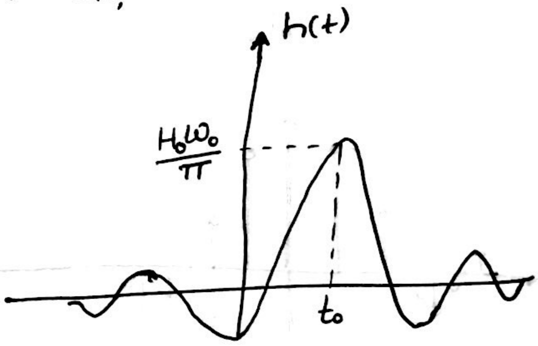
$$h_d[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(\omega)\, e^{j\omega n}\, d\omega \quad \text{——— (2)}$$

So, the desired unit sample response can be obtained from desired frequency response by using the above expressions.

For example: Consider an LPF as,



Fig(a) Magnitude Response of Ideal LPF



Fig(b): Impulse response of LPF

Now, There arises two problems with the realization of LPF.

i) Impulse response has infinite duration

ii) Non-causality and unrealisability of filter (Sinc pulse)

The impulse response is not realisable by any finite amount of delay. Therefore the resultant filter from fourier series technique is an unrealized IIR filter.

The infinite duration impulse response may be changed to a finite duration ~~impulse response may be changed to a finite duration~~ by performing the truncation procedure on the infinite series at $n = \pm M$. This is equivalent to multiplying $h_d[n]$ by a window sequence $w[n]$.

<u>For example</u>: let us consider a rectangular window which is defined as:

$$w_R[n] = \begin{cases} 1, & \text{for } n = 0, 1, 2, \cdots, (M-1) \\ 0, & \text{elsewhere} \end{cases}$$

Now, The unit sample response of the filter is obtained by multiplying $h_d[n]$ by $w_R[n]$ i.e

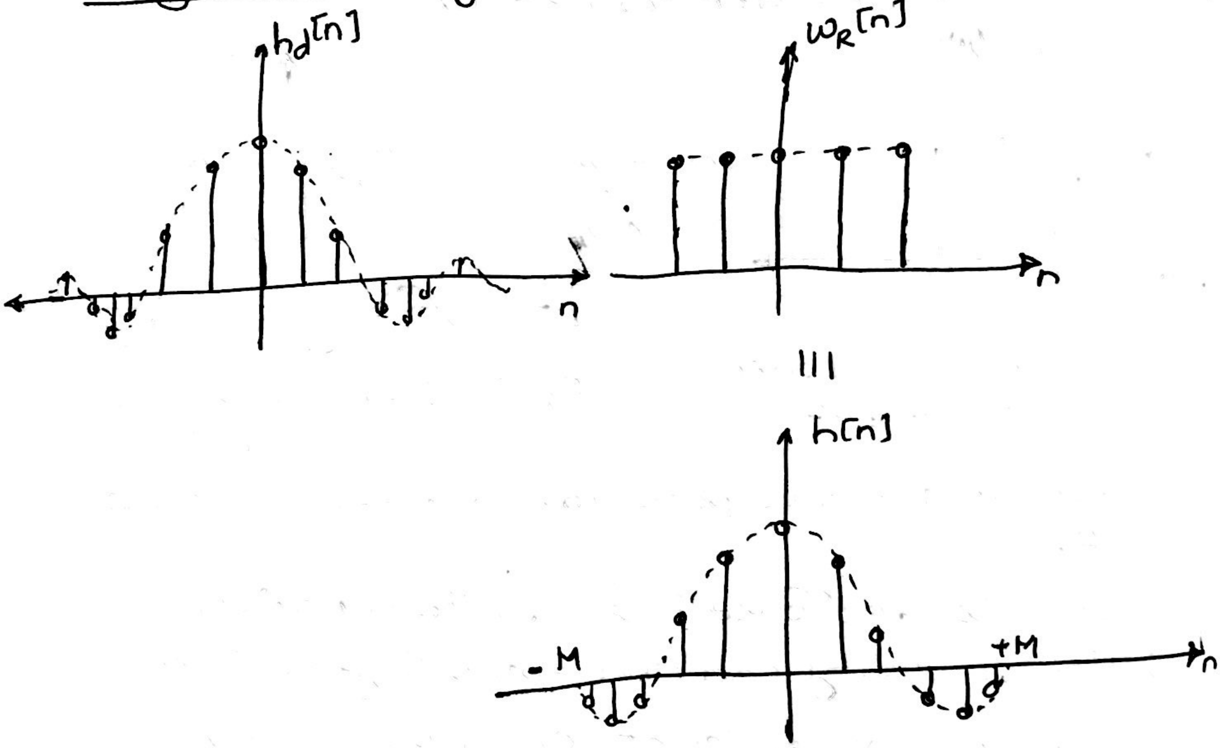
$$h[n] = h_d[n] \cdot w_R[n]$$

Then,

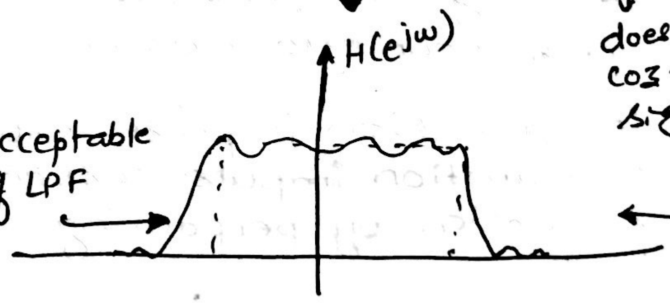
$$h[n] = \begin{cases} h_d[n], & \text{for } n = 0, 1, 2, 3, \cdots (M-1) \\ 0, & \text{otherwise.} \end{cases}$$

This is called Windowing since $w_R[n]$ acts like a window for $h_d[n]$.

<u>Diagrammatically</u>,



Inverse Fourier Transform (IFT) ↓ doesnot give ideal cos truncated signal is obtained we get ripple like



This is acceptable response of LPF

From this technique undesired oscillations are obtained which is due to the slow convergence of the fourier series nearby the points of discontinuity. These undesirable oscillations is known as Gibb's oscillations (Gibb's phenomenon).

These unwanted oscillations may be minimized by using a set of time-limited weighting functions i.e $w[n]$ known as window functions, these window function does the work of modification of fourier series Coefficients.

A major advantages of window function techniques is that existed discontinuities in $H(e^{jw})$ is converted into transition bands between values on either side of the discontinuity. The width of those transition band is dependent on the width of main-lobe $W(e^{jw})$. The second The second major advantage of the windowing technique is that ripples from the side lobes of $W(e^{jw})$ generates approximation errors for all frequencies 'w'.

There are two important desired characterstics given below:

(a) The Fourier Transform of weighting function $w[n]$ i.e $W(e^{jw})$ should possess small width of main lobe contains maximum amount of total energy as possible.

(b) The Fourier Transform of weighting function $w[n]$ i.e $W(e^{jw})$ should consist of side lobe which decreases in energy fast as frequency 'w' goes towards π.

Based on the function, we can classify the window's function into following types:

1. Rectangular window function.
2. Raised Cosine window function.
   (a) Hamming Window function.
   (b) Hanning Window function.
3. Triangular Window function (Bartlett)
4. Kaiser Window function.
(c) Blackman Window function.

ALGORITHM TO DESIGN AN IDEAL FILTER USING ANY WINDOW METHOD

① The problem gives desired frequency response $H_d(e^{j\omega})$, type of window and the order of the filter (M).

The window function are defined for the range

$$-\frac{M-1}{2} \le n \le \frac{M-1}{2} \quad \text{if } M \Rightarrow \text{odd}$$

For e.g: $M=7$, $-3 \le n \le 3$

and, $$\left[-\left(\frac{M}{2}-1\right) \le n \le \frac{M}{2}\right] \text{ OR } \left(-\frac{M}{2} \le n \le \frac{M}{2}-1\right) \quad \text{if } M \Rightarrow \text{even}$$

For e.g: $M=6$, $-2 \le n \le 3$ OR $-3 \le n \le 2$

Note: Since, sinc function has even symmetry i.e $h[-1] = h[1]$, $h[-2] = h[2]$ and so on we obtain a FIR system with finite impulse response of length 'M' using the property of sinc function. Therefore, the range of the samples is not from 0 to (M-1).

② Calculate desired impulse response using the formula,

$$h_d[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(e^{j\omega}) e^{j\omega n} \, d\omega$$

Obtain the desired filter coefficients from

$$n = 0, \pm 1, \pm 2, \ldots\ldots, \pm\left(\frac{N-1}{2}\right)$$

③ Obtain the FIR filter coefficients by multiplying $h_d[n]$ with respective window function, that is,

$$h[n] = h_d[n] \times w[n]$$

Where,

For Rectangular window,

$$w_R[n] = \begin{cases} 1, & |n| \le \frac{M-1}{2} \\ 0, & \text{otherwise} \end{cases}$$

For Triangular window (Bartlett window),

$$w_B[n] = \begin{cases} 1 - \frac{2|n|}{M-1} & ; |n| \le \frac{M-1}{2} \\ 0 & , \text{otherwise} \end{cases}$$

For Raised Cosine window, (For Non-Causal) ⇒ Change if the window is causal) Referring the for[...] befo[...]

$$w_c[n] = \alpha + (1-\alpha)\cos\left(\frac{2\pi n}{M-1}\right), \quad |n| \le \frac{M-1}{2}$$

ⓐ For Hanning window, $\alpha = 0.5$

$$w_{han}[n] = 0.5 + 0.5\cos\left(\frac{2\pi n}{M-1}\right), \quad |n| \le \left(\frac{M-1}{2}\right)$$

(b) For Hamming window, $\alpha = 0.54$

$$w_{ham}[n] = 0.54 + 0.46 \cos\left(\frac{2\pi n}{M-1}\right), \quad |n| \le \frac{M-1}{2}$$

(c) For Blackmann window

$$w_B[n] = 0.42 + 0.5 \cos\left(\frac{2\pi n}{M-1}\right) + 0.08 \cos\left(\frac{4\pi n}{M-1}\right)$$

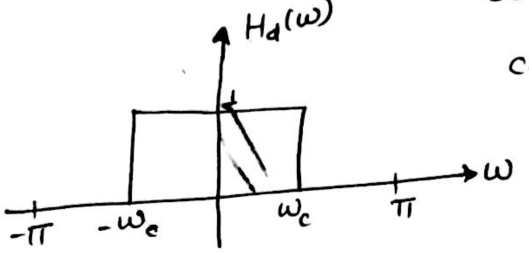
$$|n| \le \frac{M-1}{2}$$

The actual filter coefficients can be obtained for,

$$n = 0, \pm 1, \pm 2, \ldots\ldots, \pm \frac{M-1}{2}$$

Desired Impulse Response of Ideal filters

(a) For Low-Pass Filter



Consider,
magnitude = 1
cut-off frequency = ω_c

Ideal LPF allows low frequency components to pass through ($|\omega| < \omega_c$) or $-\omega_c < \omega < \omega_c$ and stops high frequency components $[\omega_c < |\omega|]$.

cut-off frequencies are those frequencies which specify the frequency ranges for which the filter allows signal to pass or stop. It is that frequency point at which magnitude transition from maximum to minimum value or minimum to maximum value occurs.

Now, Desired impulse response is,

$$h_d[n] = \text{IDTFT of } H_d(\omega)$$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} 1 \cdot e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \times \left.\frac{e^{j\omega n}}{jn}\right|_{-\omega_c}^{\omega_c}$$

$$= \frac{1}{2\pi j n}\left(e^{j\omega_c n} - e^{-j\omega_c n}\right) = \frac{1}{\pi n} \times \sin(\omega_c n)$$

$$= \frac{\omega_c}{n} \cdot \frac{\sin \omega_c n}{\omega_c n}$$

$$\boxed{\therefore h_d[n] = 2f_c \frac{\sin \omega_c n}{\omega_c n} = 2 f_c \,\text{sinc}\, \omega_c n}$$ $\because \omega_c = 2\pi$

at $n=0$, $h_d[0] = 2F_c \cdot 1 = 2f_c$ $(\because \text{sinc}\, 0 = 1)$

$n \neq 0$, $h_d[n] = 2 f_c \,\text{sinc}\, \omega_c n$

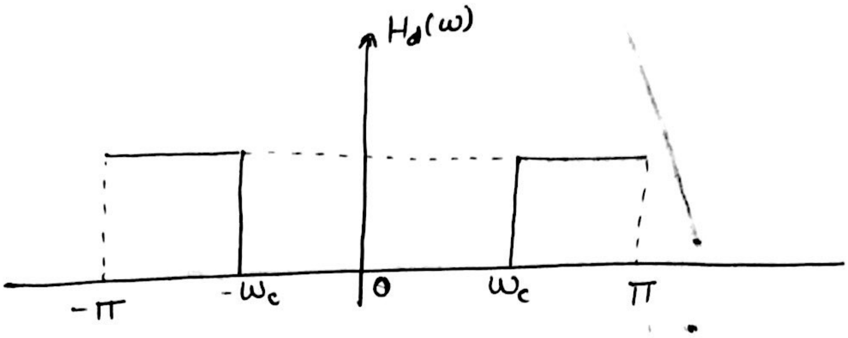
(2) $\Omega_c = 2000\pi$
$F_s = 5000$
$\omega_c = \frac{\Omega_c}{F_s}$

Note : 'f_c' should be in normalised form as

$$f_c = \frac{F_c}{F_s} = \frac{\text{cut-off frequency in Hz}}{\text{Sampling frequency in Hz}}$$

$F_c = 1000Hz$, $F_s = 5000Hz$
$f_c = \frac{F_c}{F_s} = 0.2$
$\omega_c = 2\pi f_c = 0.4\pi$

ⓑ High-Pass Filter



Low-frequency components are stopped ($|\omega| < \omega_c$) and high-frequency components are passed ($\omega_c < |\omega|$)

Now,

Desired impulse response,

$$h_d[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \left[\int_{-\pi}^{-\omega_c} 1 \cdot e^{j\omega n} d\omega + \int_{\omega_c}^{\pi} 1 \cdot e^{j\omega n} d\omega \right]$$

$$= \frac{1}{2\pi j n} \left[e^{-j\omega_c n} - e^{-j\pi n} + e^{j\pi n} - e^{j\omega_c n} \right]$$

$$= \frac{1}{2\pi j n} \left\{ \left(e^{j\pi n} - e^{-j\pi n} \right) - \left(e^{j\omega_c n} - e^{-j\omega_c n} \right) \right\}$$

$$= \frac{1}{\pi n} \left\{ \frac{e^{j\pi n} - e^{-j\pi n}}{2j} - \frac{e^{j\omega_c n} - e^{-j\omega_c n}}{2j} \right\}$$

$$= \frac{1}{\pi n} \left(\cancelto{0}{\sin \pi n} - \sin \omega_c n \right)$$

$$= -\frac{\sin \omega_c n}{\pi n} = -\frac{\omega_c}{\pi} \cdot \frac{\sin(\omega_c n)}{\omega_c n}$$

$$\boxed{\therefore h_d[n] = -2F_c \, \mathrm{sinc}(\omega_c n)} \quad , \text{for } n \neq 0$$

For $n = 0$,

$$h_d[0] = \frac{1}{2\pi} \left\{ \int_{-\pi}^{-\omega_c} 1 \cdot 1 \cdot d\omega + \int_{\omega_c}^{\pi} 1 \cdot 1 \, d\omega \right\}$$

$$= \frac{1}{2\pi}\left\{ \omega \Big|_{-\pi}^{-\omega_c} + \omega \Big|_{\omega_c}^{\pi} \right\}$$

$$= \frac{1}{2\pi}\left(-\omega_c + \pi + \pi - \omega_c \right)$$

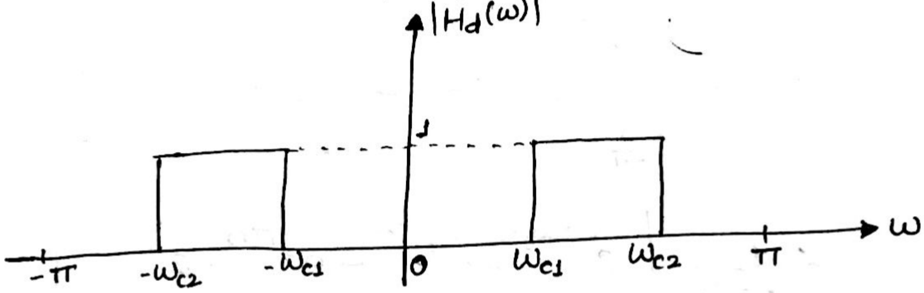
$$= \frac{1}{\pi}(\pi - \omega_c) = 1 - \frac{\omega_c}{\pi} = 1 - 2F_c$$

$$\boxed{\therefore\ h_d[0] = 1 - 2F_c}$$

© Band-Pass Filter

magnitude = 1

cut-off frequencies = ω_{c1} & ω_{c2}



Signal is passed for the range $\omega_{c1} < |\omega| < \omega_{c2}$ and & is attenuated for $|\omega| < \omega_{c1}$ and $\omega_{c2} < |\omega|$.

Now, The desired impulse response

$$h_d[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(\omega)e^{j\omega n}\,d\omega$$

$$= \frac{1}{2\pi}\left\{ \int_{-\omega_{c2}}^{-\omega_{c1}} 1\cdot e^{j\omega n}d\omega + \int_{\omega_{c1}}^{\omega_{c2}} 1\cdot e^{j\omega n}d\omega \right\}$$

at $n=0$, $h_d[0] = \frac{1}{2\pi}\left\{ \int_{-\omega_{c2}}^{-\omega_{c1}} 1\cdot d\omega + \int_{\omega_{c1}}^{\omega_{c2}} 1\cdot d\omega \right\}$

$$= \frac{1}{2\pi}\left(-\omega_{c1} + \omega_{c2} + \omega_{c2} - \omega_{c1} \right)$$

$$= \frac{1}{2\pi}\left(2\omega_{c2} - 2\omega_{c1} \right)$$

$$= \frac{1}{\pi}(\omega_{c2} - \omega_{c1}) = 2\left(F_{c2} - F_{c1}\right)$$

For, $n \neq 0$,

$$h_d[n] = \frac{1}{2\pi}\left\{ \frac{e^{j\omega n}}{jn}\Bigg|_{-\omega_{c2}}^{-\omega_{c1}} + \frac{e^{j\omega n}}{jn}\Bigg|_{\omega_{c1}}^{\omega_{c2}} \right\}$$

$$= \frac{1}{2\pi jn}\left\{ e^{-j\omega_{c1}n} - e^{-j\omega_{c2}n} + e^{j\omega_{c2}n} - e^{j\omega_{c1}n} \right\}$$

$$= \frac{1}{\pi n}\left\{ \frac{e^{j\omega_{c2}n} - e^{-j\omega_{c2}n}}{2j} - \frac{e^{j\omega_{c1}n} - e^{-j\omega_{c1}n}}{2j} \right\}$$

$$= \frac{1}{\pi n}\left\{ \sin(\omega_{c2}n) - \sin(\omega_{c1}n) \right\}$$

$$= \frac{1}{\pi n}\sin(\omega_{c2}n) - \frac{1}{\pi n}\sin(\omega_{c1}n)$$

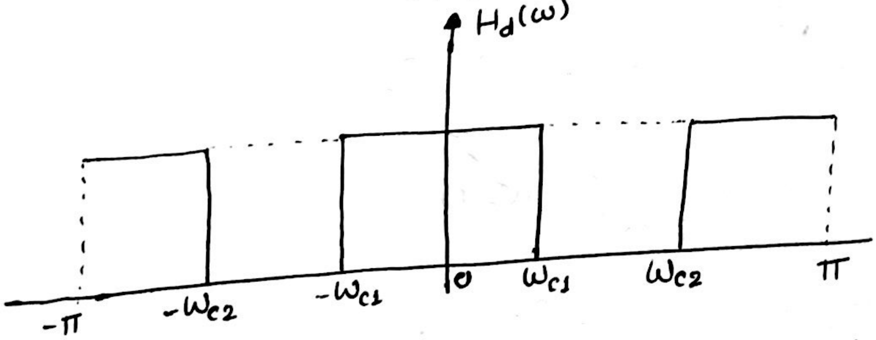
$$= \frac{\omega_{c2}}{\pi}\frac{\sin(\omega_{c2}n)}{\omega_{c2}n} - \frac{\omega_{c1}}{\pi}\cdot\frac{\sin(\omega_{c1}n)}{\omega_{c1}n}$$

$$\boxed{\therefore h_d[n] = 2F_{c2}\frac{\sin(\omega_{c2}n)}{\omega_{c2}n} - 2F_{c1}\frac{\sin(\omega_{c1}n)}{\omega_{c1}n}}$$

(d) Band-stop Filter

magnitude $= 1$

cut-off frequencies $= \omega_{c1}$ and ω_{c2}



The signal is attenuated or stopped for the frequency range, $(\omega_{c1} < |\omega| < \omega_{c2})$ and the signal is passed for the frequency range, $|\omega| < \omega_{c1}$ and $\omega_{c2} < |\omega|$

Now,
The desired impulse response,

$$h_d[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(\omega)e^{j\omega n}\,d\omega$$

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$$= \frac{1}{2\pi}\left\{\int_{-\pi}^{-\omega_{c2}} 1\cdot e^{j\omega n}\,d\omega + \int_{-\omega_{c1}}^{\omega_{c1}} 1\cdot e^{j\omega n}\,d\omega + \int_{\omega_{c2}}^{\pi} 1\cdot e^{j\omega n}\,d\omega\right\}$$

at $n=0$,

$$h_d[0] = \frac{1}{2\pi}\left\{-\omega_{c2}+\pi+\omega_{c1}+\omega_{c1}+\pi-\omega_{c2}\right\}$$

$$= \frac{1}{2\pi}\left(2\pi + 2\omega_{c1} - 2\omega_{c2}\right)$$

$$= \frac{2(\omega_{c1}-\omega_{c2})}{2\pi} = 1 + \frac{\cancel{4\pi}^{\,2}(F_{c1}-F_{c2})}{\cancel{2\pi}}$$

$$\therefore h_d[0] = 1 + 2(F_{c1}-F_{c2})$$

at $n\neq 0$,

$$h_d[n] = \frac{1}{2\pi}\left\{\left.\frac{e^{j\omega n}}{jn}\right|_{-\pi}^{-\omega_{c2}} + \left.\frac{e^{j\omega n}}{jn}\right|_{-\omega_{c1}}^{\omega_{c1}} + \left.\frac{e^{j\omega n}}{jn}\right|_{\omega_{c2}}^{\pi}\right\}$$

$$= \frac{1}{2\pi jn}\left\{e^{-j\omega_{c2}n} - e^{-j\pi n} + e^{j\omega_{c1}n} - e^{-j\omega_{c1}n} + e^{j\pi n} - e^{j\omega_{c2}n}\right\}$$

$$= \frac{1}{\pi n}\left\{\frac{e^{j\pi n}-e^{-j\pi n}}{2j} + \frac{e^{j\omega_{c1}n}-e^{-j\omega_{c1}n}}{2j} - \frac{e^{j\omega_{c2}n}-e^{-j\omega_{c2}n}}{2j}\right\}$$

$$= \frac{1}{\pi n}\left\{\cancelto{0}{\sin\pi n} + \sin\omega_{c1}n - \sin\omega_{c2}n\right\}$$

$$= \frac{1}{\pi n}\sin\omega_{c1}n - \frac{1}{\pi n}\sin\omega_{c2}n$$

$$= \frac{\omega_{c1}}{\pi}\cdot\frac{\sin\omega_{c1}n}{\omega_{c1}n} - \frac{\omega_{c2}}{\pi}\cdot\frac{\sin\omega_{c2}n}{\omega_{c2}n}$$

$$\boxed{\therefore h_d[n] = 2F_{c1}\,\mathrm{sinc}\,\omega_{c1}n - 2F_{c2}\,\mathrm{sinc}\,\omega_{c2}n}$$

#

Desired Impulse Response of different Ideal Filters

S.NO.	Filter Types	Desired Impulse Response $h_d[n]$ When $n \neq 0$	Desired Impulse Response, $h_d[n]$, when, $n=0$
1.	LPF	$h_d[n] = 2F_c \, Sin(\omega_c n)$	$h_d[0] = 2F_c$
2.	HPF	$h_d[n] = -2F_c \, Sin(\omega_c n)$	$h_d[0] = 1 - 2F_c$
3.	BPF	$h_d[n] = 2F_{c2} \, Sinc(\omega_{c2} n) - 2F_{c1} \, Sinc(\omega_{c1} n)$	$h_d[0] = 2(F_{c2} - F_{c1})$
4.	BSF	$h_d[n] = 2F_{c1} \, Sinc(\omega_{c1} n) - 2F_{c2} \, Sinc(\omega_{c2} n)$	$h[0] = 1 + 2(F_{c1} - F_{c2})$

Design a low-Pass Linear phase FIR filter with 11 Coefficients for the following specification with cut-off frequency = 0.25kHz and sampling frequency = 1kHz.

→ Given that,

Cut-off frequency, $f_c = 0.25kHz$

Sampling frequency, $F_{SF} = 1kHz$

so,

Normalized cut-off frequency, $F_c = \frac{f_c}{F_{SF}} = \frac{0.25 \times \cancel{10^3}}{1 \times \cancel{10^3}}$

$= 0.25$

and,

Normalized cut-off frequency, $\omega_c = 2\pi F_c$

$= 2\pi \times 0.25 = 0.5\pi$

order of filter, $M = 11 \Rightarrow -\frac{M-1}{2} \le n \le \frac{M-1}{2}$

For Ideal LPF, $\Rightarrow -5 \le n \le 5$

The desired Impulse response is,

$$h_d[n] = \begin{cases} 2F_c \dfrac{\sin(n\omega_c)}{n\omega_c} & , n \neq 0 \\ 2F_c & , n = 0 \end{cases}$$

Now, Computing the coefficients,

$n=0$, $h_d[0] = 2F_c = 2 \times 0.25 = 0.5$

$n=1$, $h_d[1] = 2 \times 0.25 \frac{\sin(1 \times 0.5 \times 180)}{0.5 \times 3.14 \times 1} = 0.3184 = h_d[-1]$

$n=2$, $h_d[2] = 2 \times 0.25 \frac{\sin(0.5 \times 180 \times 2)}{0.5 \times 3.14 \times 2} = 0 = h_d[-2]$

$n=3$, $h_d[3] = 2 \times 0.25 \frac{\sin(0.5 \times 180 \times 3)}{0.5 \times 3.14 \times 3} = -0.1062 = h_d[-3]$

$n=4$, $h_d[4] = 2 \times 0.25 \frac{\sin(0.5 \times 180 \times 4)}{0.5 \times 3.14 \times 4} = 0 = h_d[-4]$

$n=5$, $h_d[5] = 2 \times 0.25 \frac{\sin(0.5 \times 180 \times 5)}{0.5 \times 3.14 \times 5} = 0.06369 = h_d[-5]$

Using rectangular window,

$$w_R[n] = \begin{cases} 1, & -5 \le n \le 5 \\ 0, & \text{elsewhere} \end{cases}$$

So, overall impulse Response of the FIR filter is

$$h[n] = h_d[n] \cdot w_R[n]$$

$$= h_d[n] \quad -5 \le n \le 5$$

$\therefore h[0] = h_d[0] = 0.5$

$h[1] = h_d[1] = 0.3184 = h[-1]$

$h[2] = h_d[2] = 0 = h[-2]$

$h[3] = h_d[3] = -0.1062 = h[-3]$

$h[4] = h_d[4] = 0 = h[-4]$

$h[5] = h_d[5] = 0.06369 = h[-5]$

Here,

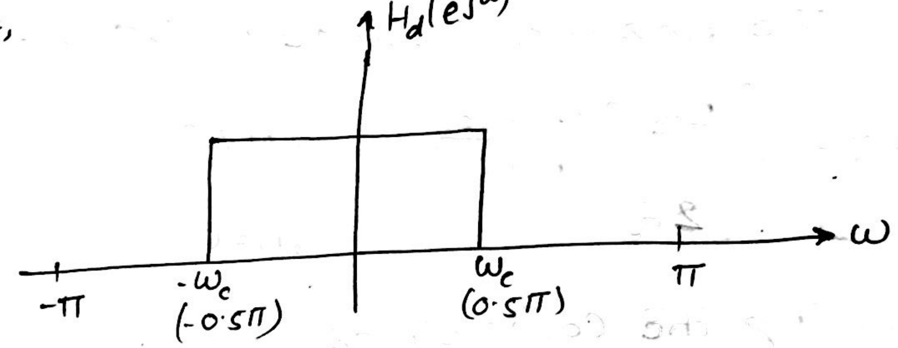


Fig: Desired Frequency Response

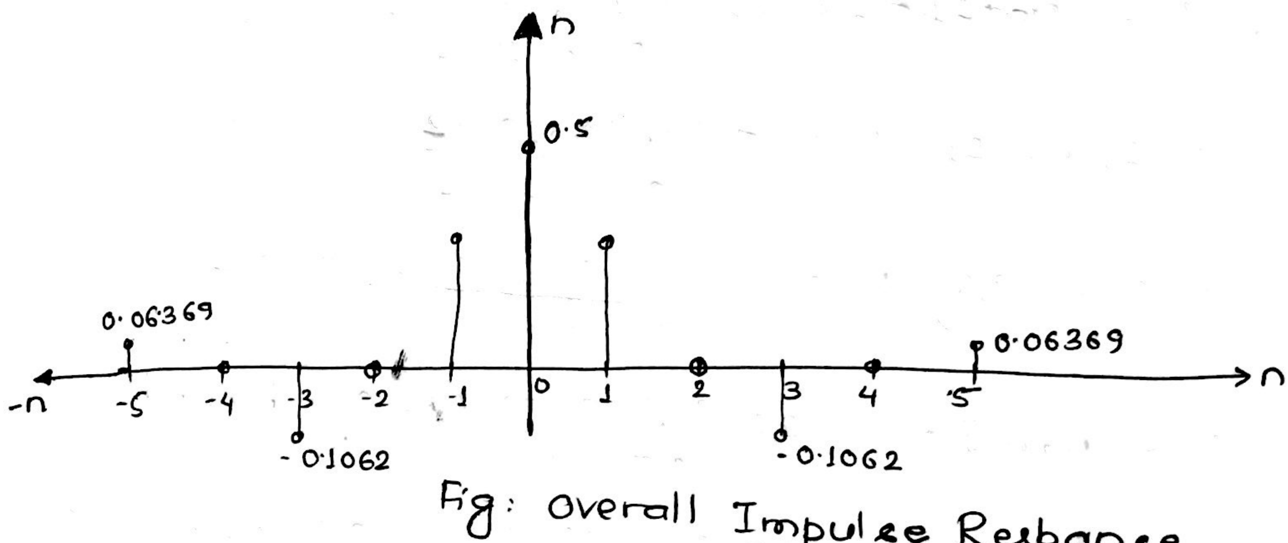


Fig: Overall Impulse Response

Note : Do same for all 4-Windows.

Design a FIR system to meet the following specification.

passband edge = 2kHz

stopband edge = 5kHz

Stopband Attenuation = 42dB

Sampling frequency, F_{SF} = 20kHz

→ Given that,

Sampling frequency, $F_{SF} = 20kHz$

passband edge frequency, $f_p = 2kHz$

∴ Normalised passband edge frequency, $F_P = \frac{f_P}{F_{SF}}$

$$\text{or, } F_P = \frac{2\times10^3}{20\times10^3} = 0.1$$

Stopband edge frequency, $f_s = 5kHz$

∴ Normalised stopband edge frequency, $F_S = \frac{f_s}{F_{SF}}$

$$\therefore F_S = \frac{5\times10^3}{20\times10^3} = \frac{1}{4} = 0.25.$$

Stopband Attenuation, $A_S = 42dB$

i.e Atleast 42 dB attenuation is required in stopband it is obvious that Hanning window is used because it provides minimum attenuation of 44dB, which is given by,

$$\omega_{Han}[n] = \begin{cases} 0.5 + 0.5\cos\left(\frac{2\pi n}{M-1}\right), & -\frac{M-1}{2} \le n \le \frac{M+1}{2} \\ 0, & \text{otherwise.} \end{cases}$$

Where M = order of the filter and is to be determined in this case (∵ not given)

But We know that, For Hanning window,

Window length, $M = \frac{3.1}{\Delta f}$

$$\text{or, } M = \frac{3.1}{F_S - F_P} = \frac{3.1}{0.25-0.1} = 20.667 = 21 \text{(odd)}$$

We wish to design symmetric FIR filter with length odd Then, It must satisfy the eqn.

$$h[n] = h[M-1-n] = h[20-n]$$

so, The window function is defined as,

$$w_{Han}[n] = \begin{cases} 0.5 + 0.5\cos\left(\frac{2\pi n}{21-1}\right) & , -\frac{M-1}{2} \le n \le \frac{M+1}{2} \\ 0 & , \text{elsewhere} \end{cases}$$

$$= \begin{cases} 0.5 + 0.5\cos\left(\frac{\pi n}{10}\right) & , -10 \le n \le 10 \\ 0 & , \text{elsewhere} \end{cases}$$

Since, the specification implies to design FIR LPF and we assume the ideal LPF, whose desired impulse Response is,

$$h_d[n] = \begin{cases} 2F_c \frac{\sin(\omega_c n)}{\omega_c n} & , n \neq 0 \\ 2F_c & , n = 0 \end{cases}$$

Where, $F_c = \frac{F_s + F_p}{2} = \frac{0.25+0.1}{2} = 0.175$

and, $\omega_c = 2\pi F_c = 2\pi \times 0.175 = 0.35\pi$

So,

$$h_d[n] = \begin{cases} 2\times 0.175 \frac{\sin(0.35\pi n)}{0.35 \times 3.14 n} & , n \neq 0 \\ 2 \times 0.175 & , n = 0 \end{cases}$$

* Calculator in Radian mode

$$= \begin{cases} \cancel{0.35} \frac{\sin(0.35\pi n)}{\cancel{0.35} \times \pi n} & , n \neq 0\ (-10 \le n \le 10) \\ 0.35 & , n = 0 \end{cases}$$

Now, The overall response of the FIR filter is

$$h[n] = h_d[n]\, w[n]$$

$$\therefore h[n] = \begin{cases} \frac{\sin(0.35\pi n)}{\pi n} \times 0.5 + 0.5\cos\left(\frac{\pi n}{10}\right) & , -10 \le n \le 10 \\ 0.35 \times 0.5 + 0.5\cos\left(\frac{\pi n}{10}\right) & , n = 0 \end{cases}$$

$\therefore$ =

So, $h[n] = \{ \underset{-10}{0}, \underset{-9}{-3.93\times10^{-4}}, \underset{-8}{2.23\times10^{-3}}, \underset{-7}{9.25\times10^{-3}}, \underset{-6}{5.66\times10^{-3}}, \underset{-5}{-0.022}, \underset{-4}{-0.04},$

$\underset{-3}{-0.0132}, \underset{-2}{0.116}, \underset{-1}{0.276}, \underset{\uparrow 0}{0.35}, \underset{1}{0.276}, \underset{2}{0.116}, \underset{3}{-0.0132}, \underset{4}{-0.04},$

$\underset{5}{-0.022}, \underset{6}{-5.66\times10^{-3}}, \underset{7}{9.25\times10^{-3}}, \underset{8}{2.23\times10^{-3}}, \underset{9}{-3.93\times10^{-4}}, \underset{10}{0} \}$

Determine the FIR filter length and cut-off frequency to be used in the design equation for a LPF with the following specifications,

Passband : 0 - 1850 Hz
Stopband : 2150 - 4000 Hz
Stopband attenuation : 20 dB
Passband ripple : 1 dB
Sampling rate : 8000 Hz.

→ Solution,

From the given specification,

$f_p = 1850\,Hz$

$f_s = 2150\,Hz$

So,
The normalised transition width,

$$\Delta f = \frac{f_s - f_p}{F_{SF}} = \frac{2150-1850}{8000} = 0.0375$$

As, stopband attenuation, $A_s = 20\,dB$, Rectangular window selection would satisfy the design requirement having stopband attenuation of 21 dB. Since, order of the filter is not given, we determine this first. But, we know that, length of filter, $M = \frac{0.9}{\Delta f}$ (For rectangular window)

$$\text{or, } M = \frac{0.9}{F_s - F_p} = \frac{0.9}{0.0375} = 24$$

But, we wish to design symmetric FIR filter with odd length so, $M = 25$

∴ The length varies as,

$$-\frac{M-1}{2} \le n \le \frac{M-1}{2} \Rightarrow -12 \le n \le 12$$

Now,
for the ideal LPF, the desired impulse response is

$$h_d[n] = \begin{cases} 2F_c \dfrac{\sin(\omega_c n)}{\omega_c n} & , -12 \le n \le 12 \ , n \ne 0 \\ 2F_c & , n = 0 \end{cases}$$

Where, $\omega_c = 2\pi F_c$

$$= \cancel{2}\pi \frac{F_p + F_s}{\cancel{2}}$$

$$= \pi(F_p + F_s)$$

$$= \pi \times \left(\frac{f_p}{F_{SF}} + \frac{f_s}{F_{SF}}\right) \leftarrow \text{normalised frequency}$$

$$F_c = \frac{F_p + F_s}{2} = \frac{1}{2}\left(\frac{f_p}{F_{SF}} + \frac{f_s}{F_{SF}}\right) = \frac{1}{2}\,\frac{f_p + f_s}{F_{SF}} = 0.2$$

$$= \pi \, \frac{1850 + 2150}{8000} = 0.5\pi$$

$$\Rightarrow F_c = \frac{\omega_c}{2\pi} = \frac{0.5\cancel{\pi}}{2\cancel{\pi}} = 0.25$$

$$\therefore h_d[n] = \begin{cases} \cancel{2 \times 0.25} \dfrac{\sin(0.5\pi n)}{\cancel{0.5}\pi n} & , -12 \le n \le 12 \\ 2 \times 0.25 & , n \ne 0 \end{cases}$$

$$= \begin{cases} \dfrac{\sin(0.5\pi n)}{\pi n} & , -12 \le n \le 12 \\ 0.5 & , n = 0 \end{cases}$$

Similarly,

$$\omega_R[n] = \begin{cases} 1 & , -12 \le n \le 12 \\ 0 & , \text{otherwise.} \end{cases}$$

So, The impulse response of FIR filter is

$$h[n] = h_d[n]\,\omega_R[n]$$

$$\therefore h[n] = \begin{cases} \dfrac{\sin(0.5\pi n)}{\pi n} \times 1 & , -12 \le n \le 12 \\ 0.5 \times 1 & , n = 0 \end{cases}$$

#

Example 2: Design the symmetric FIR-LPF for which the desired frequency response is expressed as

$$H_d(\omega) = \begin{cases} e^{-j\omega\tau} & , |\omega| \le \omega_c \\ 0 & , \text{elsewhere} \end{cases}$$

The length of the filter should be 7 and $\omega_c = 1$ radians/sample. Make use of rectangular window.

Given that,

length of the filter, $M = 7$

cut-off frequency, $\omega_c = 1$ rad/sample

Then,

Desired frequency response,

$$H_d(\omega) = \begin{cases} e^{-j\omega\tau} & , |\omega| \le 1 \text{ or } (-1 \le \omega \le 1) \\ 0 & , \text{elsewhere.} \end{cases}$$

Now, for $n \ne \tau$,

The desired unit sample response is given by,

$$h_d[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(\omega)e^{j\omega n}\,d\omega = \frac{1}{2\pi}\int_{-1}^{1} e^{-j\omega\tau}\cdot e^{j\omega n}\,d\omega$$

$$= \frac{1}{2\pi}\int_{-1}^{1} e^{j\omega(n-\tau)}\,d\omega = \frac{1}{2\pi}\times \left.\frac{e^{j\omega(n-\tau)}}{j(n-\tau)}\right|_{-1}^{1}$$

$$= \frac{1}{2\pi j(n-\tau)}\times\left\{e^{j(n-\tau)} - e^{-j(n-\tau)}\right\}$$

$$= \frac{1}{\pi(n-\tau)}\times\frac{e^{j(n-\tau)} - e^{-j(n-\tau)}}{2j}$$

$$= \frac{1}{\pi(n-\tau)}\times \sin(n-\tau)$$

$$\therefore h_d[n] = \frac{\sin(n-\tau)}{\pi(n-\tau)} \quad \text{for } n \ne \tau$$

Similarly, $n = \tau$,

$$h_d[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(\omega)e^{j\omega n}\,d\omega = \frac{1}{2\pi}\int_{-1}^{1} e^{-j\omega\tau}\cdot e^{j\omega\tau}\,d\omega$$

$$= \frac{1}{2\pi}\times \omega\Big|_{-1}^{1} = \frac{1}{\cancel{2}\pi}\times \cancel{2} = \frac{1}{\pi} = \frac{1}{\pi}$$

So,
$$h_d[n] = \begin{cases} \dfrac{\sin(n-\tau)}{\pi(n-\tau)} & \text{for } n \neq \tau \\ \dfrac{1}{\pi} & \text{for } n = \tau \end{cases} \quad \text{———— (1)}$$

Since, length of the filter, $M = 7$ i.e odd and we always wish to design symmetric linear phase FIR filter (in this q.n it is asked to do so). So, for this the condition

$$h[n] = h[M-1-n] \quad \text{must be satisfied.}$$

$$\Rightarrow h_d[n] = h_d[M-1-n]$$

Using this condition on eq^n (1) yields,

$$\frac{\sin(n-\tau)}{\pi(n-\tau)} = \frac{\sin\{(M-1-n)-\tau\}}{\pi\{(M-1-n)-\tau\}}$$ i.e replacing ($n \to M-1-n$) on eq^n (1)

$$\text{or, } \frac{\sin(n-\tau)}{\pi(n-\tau)} = \frac{\sin[-(-M+1+n+\tau)]}{\pi[-(-M+1+n+\tau)]}$$

$$\text{or, } \frac{\sin(n-\tau)}{\pi(n-\tau)} = \frac{\cancel{-}\sin(-M+1+n+\tau)}{\cancel{-}\pi(-M+1+n+\tau)} \quad \because \sin(-\theta) = \sin\theta$$

$$\text{or, } \frac{\sin(n-\tau)}{\pi(n-\tau)} = \frac{\sin(-M+1+n+\tau)}{\pi(-M+1+n+\tau)}$$

The above condition will be satisfied if,

$$\cancel{n} - \tau = -M + 1 + \cancel{n} + \tau$$

$$\text{or, } M-1 = 2\tau$$

$$\therefore \tau = \frac{M-1}{2} = \frac{7-1}{2} = \frac{6}{2} = 3$$

Hence, eq^n (1) with this value becomes,

$$h_d[n] = \begin{cases} \dfrac{\sin(n-3)}{\pi(n-3)} & \text{for } n \neq 3 \\ \dfrac{1}{\pi} & \text{for } n = 3 \end{cases} \quad \text{———— (2)}$$

This implies eq^n (2) have infinite values but according to the question length of filter, $M = 7$ so the variation of 'n' would be as,

$$\underbrace{0, 1, 2}_{n \neq 3},\ \overset{n=3}{3},\ \underbrace{4, 5, 6}_{n \neq 3} \quad (\text{as per eq}^n \text{ (2)})$$

$$\Rightarrow 0 \leq n \leq 6$$

Since, rectangular window is to be used (of length 7)

$$w_R[n] = \begin{cases} 1, & 0 \le n \le 6 \\ 0, & \text{elsewhere} \end{cases}$$

also, the filter is symmetric, it must satisfy the condition

$$h[n] = h[M-1-n] = h[7-1-n] = h[6-n]$$

Now,

For $n = 0$, $h[0] = h[6]$

$h[1] = h[5]$

$h[2] = h[4]$

Finally, the impulse response is calculated as

$$h[n] = h_d[n] \cdot w_R[n].$$

Preparing the table,

S.No.	n.	$h_d[n] = \frac{\sin(n-3)}{\pi(n-3)}$ $n \neq 3$	$w_R[n]$	$h[n] = h_d[n] \cdot w_R[n]$
1	0	0.01497	1	0.01497
2	1	0.14472	1	0.14472
3	2	0.26785	1	0.26785
4	3	0.31831	1	0.31831
5	4	0.26785 = h[2]	1	0.26785
6	5	0.14472 = h[1]	1	0.14472
7	6	0.01497 = h[0]	1	0.01497

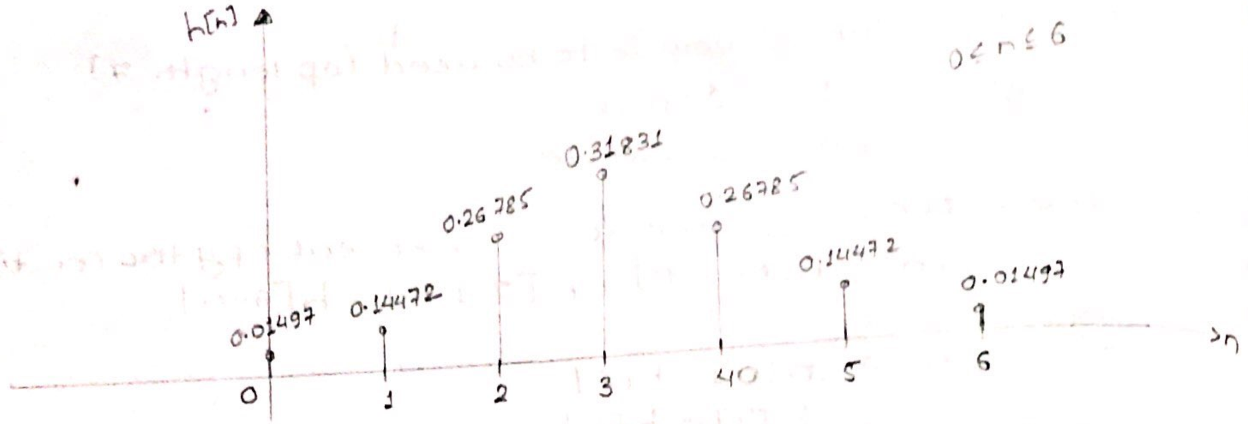


Fig: Symmetric FIR LPF (M = 7) Using Rectangular window

Example: 3 Design the FIR filter of example 2, using Hanning window function.

→ For Hanning Window function, We have, ($0 \le n \le M-1$)

$$w_{han}[n] = 0.5 - 0.5\cos\left(\frac{2\pi n}{M-1}\right), \quad 0 \le n \le 6$$

$M = 7$

⇓

Window is Causal in nature. So, the formula is used

$$= 0.5 - 0.5\cos\left(\frac{2\pi n}{6}\right)$$

$$= 0.5 - 0.5\cos\left(\frac{\pi n}{3}\right)$$

Preparing the table,

S.NO.	n	$h_d[n] = \frac{\sin(n-3)}{\pi(n-3)}$, $n \ne 3$	$w_{han}[n] = 0.5 - 0.5\cos\left(\frac{\pi n}{3}\right)$	$h[n] = h_d[n] \cdot w_{han}[n]$
1	0	$h_d[0] = 0.01497$	$w_{han}[0]$ 0	$h[0] = 0$
2	1	$h_d[1] = h_d[5] = 0.14472$	$w_{han}[1] = 1/4$	$h[1] = 0.03618$
3	2	$h_d[2] = h_d[4] = 0.26785$	$w_{han}[2] = 3/4$	$h[2] = 0.20089$
4	3 (n=3)	$h_d[3] = 0.31831$	$w_{han}[3] = 1$	$h[3] = 0.31831$
5	4	$h_d[4] = 0.26785$	$w_{han}[4] = 3/4$	$h[4] = 0.20089$
6	5	$h_d[5] = 0.14472$	$w_{han}[5] = 1/4$	$h[5] = 0.03618$
7	6	$h_d[6] = 0.01497$	$w_{han}[6] = 0$	$h[6] = 0$

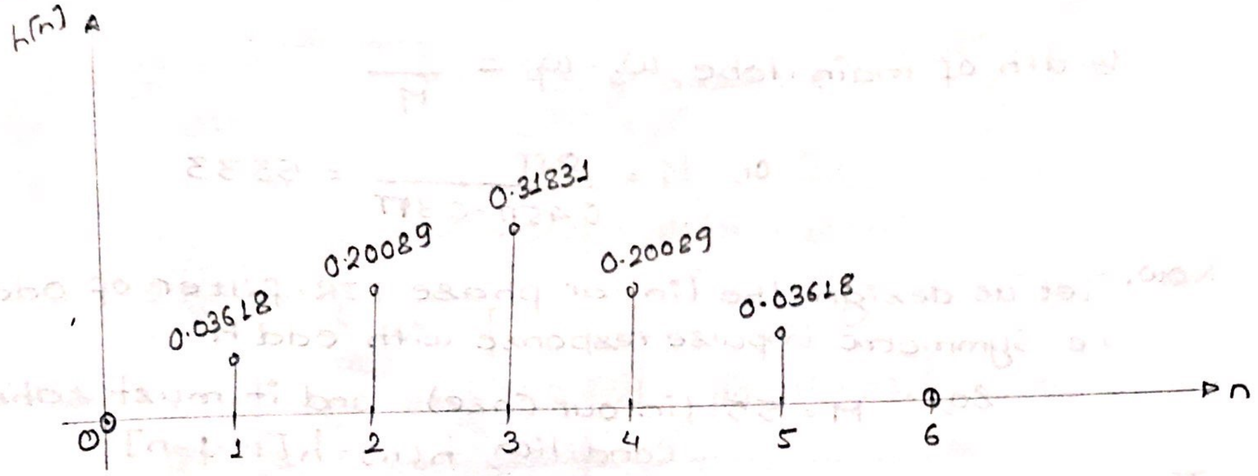


Fig: Symmetric FIR filter using Hanning window.

Example 4: Design a Low-pass digital Filter that will have -3dB cut-off frequency at 30π rad/sec and an attenuation of 50dB at 45π rad/sec. The filter is required to have a linear phase and system uses a sampling rate of 100 samples/second.

→ Given that,

3-dB cut-off frequency, $\Omega_c = 30\pi$ rad/sec (in analog domain) $= \Omega_p$

Sampling frequency, F_{SF} = 100 samples/second

Normalising the cut-off frequency (i.e converting analog domain frequency into digital domain) as,

$$\omega_c = \frac{\Omega_c}{F_{SF}} = \frac{30\pi}{100} = 0.3\pi \text{ rad/sample} = \omega_p$$

Stop-band Attenuation, $A_s = 50$dB

Stop-band edge frequency, $\Omega_s = 45\pi$ rad/sec

In digital domain,

stop-band edge frequency, $\omega_s = \frac{\Omega_s}{F_{SF}} = \frac{45\pi}{100} = 0.45$ rad/sample.

Here, window function is not mentioned directly. In such cases stop-band attenuation (A_s) is used as reference inorder to decide the appropriate window function. ~~Also,~~ Minimum stopband attenuation is given as 50dB. But, we know that hamming window provides -53dB of stop ~~pass~~band attenuation. Therefore, we should use hamming window to obtain the desired attenuation of 50dB.

Also, order of the filter is to be determined, M = ?

For, Hamming window,

~~Main lobe width~~

Width of main-lobe, $\omega_s - \omega_p = \frac{8\pi}{M}$

$$\text{or, } M = \frac{8\pi}{0.45\pi - 0.3\pi} = 53.33$$

Now, let us design the linear phase FIR-filter of odd len, i.e Symmetric impulse response with odd M

so, $M = 55$ (in our case) and it must satisfy condition, $h[n] = h[M-1-n]$

Then,

For Hamming window, We have, [causal], $(0 \le n \le M-1)$

$$\omega_{Ham}[n] = 0.54 - 0.46\cos\left(\frac{2\pi n}{M-1}\right)$$

$$= 0.54 - 0.46\cos\left(\frac{2\pi n}{55-1}\right)$$

$$= 0.54 - 0.46\cos\left(\frac{2\pi n}{56}\right)$$

$$\therefore \omega_{Ham}[n] = 0.54 - 0.46\cos\left(\frac{\pi n}{27}\right) \quad \text{--- (1)}$$

Now, In order to calculate the desired impulse respons $h_d[n]$, Assume desired frequency response as

$$H_d(\omega) = \begin{cases} e^{-j\omega\tau} & , |\omega| \le \omega_c \\ 0 & , \text{elsewhere} \end{cases}$$

(τ = phase delay = d in our case)

Since, it is linear pha

$\tau = \frac{M-1}{2} = \frac{55-1}{2} = 27$

$\omega_c = 0.3\pi$

For our case,

$$H_d(\omega) = \begin{cases} e^{-j27\omega} & , |\omega| \le 0.3\pi \\ 0 & , \text{elsewhere} \end{cases}$$

Then,

The desired unit sample response is given by,

$$h_d[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(\omega)e^{j\omega n}\,d\omega$$

$$= \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} H_d(\omega)e^{j\omega n}\,d\omega$$

For, $n \ne 27$,

$$= \frac{1}{2\pi}\int_{-0.3\pi}^{0.3\pi} e^{-j27\omega}\cdot e^{j\omega n}\,d\omega$$

$$= \frac{1}{2\pi}\int_{-0.3\pi}^{0.3\pi} e^{j\omega(n-27)}\, d\omega = \frac{1}{2\pi} \times \frac{1}{j(n-27)} \times e^{j\omega(n-27)}\Big|_{-0.3\pi}^{0.3\pi}$$

$$= \frac{1}{2\pi j(n-27)} \times \left\{ e^{j[0.3\pi(n-27)]} - e^{-j[0.3\pi(n-27)]} \right\}$$

$$= \frac{1}{\pi(n-27)} \times \frac{e^{j[0.3\pi(n-27)]} - e^{-j[0.3\pi(n-27)]}}{2j}$$

$$= \frac{1}{\pi(n-27)} \times \sin[0.3\pi(n-27)]$$

$$\therefore h_d[n] = \frac{1}{\pi} \times \frac{\sin[0.3\pi(n-27)]}{(n-27)} \quad \text{---} \; (2)$$

Similarly, For $n = 27$

The desired unit impulse response is,

$$h_d[n] = \frac{1}{2\pi}\int_{-0.3\pi}^{0.3\pi} e^{-j27\omega} \cdot e^{j27\omega}\, d\omega = \frac{1}{2\pi}\int_{-0.3\pi}^{0.3\pi} d\omega$$

$$= \frac{1}{2\pi} \times [0.3\pi - (-0.3\pi)] = \frac{0.6\cancel{\pi}}{2\cancel{\pi}} = 0.3 \quad \text{---} \; (3)$$

Finally,

The impulse response of symmetric FIR filter with odd length ($M = 55$) is given by,

$$h[n] = h_d[n] \times \omega_{Ham}[n]$$

$$h_d[n] = \begin{cases} \left\{\frac{1}{\pi} \times \frac{\sin[0.3\pi(n-27)]}{(n-27)}\right\} \times \left\{0.54 - 0.46\cos\left(\frac{\pi n}{27}\right)\right\}, & n \neq 27 \\ 0.3 \times \left\{0.54 - 0.46\cos\left(\frac{\pi n}{27}\right)\right\}, & n = 27 \end{cases}$$

#

Example 5 : Design a normalised linear phase FIR filter having the phase delay of $\tau = 4$ and at least 40dB attenuation in stopband. Also, obtain the magnitude/frequency response of the filter.

$\longrightarrow$ Given that,

Stop band attenuation, $A_s = 40dB$

Now,

At least 40dB attenuation is required in stopband, it is obvious that Hanning window is used because it provides minimum attenuation of 44dB, which is given by,

$$\omega_{Han}[n] = 0.5 - 0.5\cos\left(\frac{2\pi n}{M-1}\right) \quad , \quad 0 \le n \le \frac{M-1}{2} \quad \text{(Causal window)}$$

But, phase delay, $\tau = 4$ (α is used as notation as well)

We prefer to design linear phase filter, so it must satisfy the condition,

$$\tau = \frac{M-1}{2}$$

$$\text{or, } 4 = \frac{M-1}{2} \Rightarrow M-1 = 8 \quad \therefore M = 9$$

$\therefore$ order of filter, $M = 9$

$\therefore n \Rightarrow 0, 1, 2, 3, 4, 5, 6, 7, 8$

$$\therefore \omega_{Han}[n] = 0.5 - 0.5\cos\left(\frac{2\pi n}{9-1}\right) \quad , \quad (0 \le n \le M-1)$$

$$= 0.5 - 0.5\cos\left(\frac{\pi n}{4}\right) \quad \text{——— (1)}$$

Since, normalised linear phase FIR filter is to be designed, we assume the desired frequency response as,

$$H_d(\omega) = \begin{cases} e^{-j\omega\tau} & , -1 \le \omega \le 1 \leftarrow \text{normalised value, } \omega_c = 1 \\ 0 & , \text{elsewhere} \end{cases}$$

But, $\tau = 4$

$$\therefore H_d(\omega) = \begin{cases} e^{-j\omega 4} & , -1 \le \omega \le 1 \\ 0 & , \text{elsewhere} \end{cases}$$

The desired impulse response is then given as,

For $n \ne 4$,
$$h_d[n] = \frac{1}{2\pi}\int_{-1}^{1} e^{-j4\omega} \cdot e^{j\omega n}\, d\omega$$

or, $h_d[n] = \frac{1}{2\pi}\int_{-1}^{1} e^{j\omega(n-4)}\, d\omega$

$$= \frac{1}{2\pi} \times \frac{e^{j\omega(n-4)}}{j(n-4)}\Bigg|_{-1}^{1}$$

$$\therefore h_d[n] = \frac{\sin[(n-4)]}{\pi(n-4)}, \quad \text{for } n \neq 4 \qquad \text{— (2)}$$

similarly,
For $n=4$, The desired impulse response is,

$$h_d[n] = \frac{1}{2\pi}\int_{-1}^{1} e^{-j4\omega} \cdot e^{j4\omega}\, d\omega$$

$$= \frac{1}{2\pi} \times 2 = \frac{1}{\pi} \qquad \text{— (3)}$$

So, The impulse response of ~~symmetrical~~ FIR filter is,

$$h[n] = h_d[n] \times W_{Han}[n]$$

$$h[n] = \begin{cases} \left\{\frac{\sin[(n-4)]}{\pi(n-4)}\right\} \times \left\{0.5 - 0.5\cos\left(\frac{\pi n}{4}\right)\right\}, & n \neq 4 \\ \frac{1}{\pi}\left\{0.5 - 0.5\cos\left(\frac{\pi n}{4}\right)\right\} & , n = 4 \end{cases} \qquad \text{— (4)}$$

Here, order of the filter is $M = 9$ (odd) so we choose to design a symmetric filter which satisfies the condition $h[n] = h[M-1-n]$. whose magnitude is given by,

$$|H(\omega)| = h\left(\frac{M-1}{2}\right) + 2\sum_{n=1}^{\frac{M-1}{2}} h\left(\frac{M-1}{2} - n\right)\cos\omega n$$

$$= h(4) + 2\sum_{n=1}^{4} h(4-n)\cos\omega n$$

$$= [h(4) + 2h(3)\cos\omega + 2h(2)\cos 2\omega + 2h(1)\cos 3\omega + 2h(0)\cos 4\omega)$$

But, $h[n] = h[M-1-n] = h[9-1-n] = h[8-n]$

so, putting the value on eqⁿ ④ gives,

$h[0] = h[8] = 0$

$h[1] = h[7] = 2.19 \times 10^{-3}$

$h[2] = 0.072 = h[6]$

$h[3] = 0.228 = h[5]$

$h[4] = 0.318$

so,

$$|H(e^{j\omega})| = 0.318 + 2\times 0.228\cos\omega + 2\times 0.072\cos 2\omega + 2\times 2.19\times 10^{-3}\cos 3\omega + 2\times 0\times\cos 4\omega$$

$$= 0.318 + 0.456\cos\omega + 0.144\cos 2\omega + 4.38\times 10^{-3}\cos 3\omega$$

Example 6: The desired response of a LPF is,

$$H_d(e^{j\omega}) = \begin{cases} e^{-j\omega 3}, & -\frac{\pi}{8} \le \omega \le \frac{\pi}{8} \\ 0, & \frac{\pi}{8} \le |\omega| \le \pi \end{cases}$$

Determine $H(e^{j\omega})$ and $|H(e^{j\omega})|$ for $N = 11$ using

1. Rectangular window
2. Hamming window
3. Hanning window
4. Blackmann window

$H(e^{j\omega}) \rightarrow H(z)$

In such quest we start with Non-Causal ran

Given that,

Step 1:

$$H_d(e^{j\omega}) = \begin{cases} e^{-j\omega 3}, & -\frac{\pi}{8} \le \omega \le \frac{\pi}{8} \\ 0, & \frac{\pi}{8} \le |\omega| \le \pi \end{cases}$$

$\tau = 3$

Filter order, $M = N = 11$. We always wish to design th linear phase filter. Consider symmetric FIR filter wi length odd (as given).

Now,

Step 2: The desired impulse response is,

For $n \neq 3$

$$h_d[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(e^{j\omega}).\, e^{j\omega n}\, d\omega$$

$$= \frac{1}{2\pi}\int_{-\pi/8}^{\pi/8} e^{-j3\omega} \cdot e^{j\omega n}\, d\omega = \frac{1}{2\pi}\int_{-\pi/8}^{\pi/8} e^{j\omega(n-3)}\, d\omega$$

$$= \frac{1}{2\pi} \times \frac{1}{j(n-3)} \times e^{j\omega(n-3)} \Big|_{-\pi/8}^{\pi/8}$$

$$\therefore h_d[n] = \frac{\sin\left[\frac{\pi}{8}(n-3)\right]}{\pi(n-3)}, \quad \text{For } n \neq 3$$

Similarly,

For $n = 3$,

$$h_d[n] = \frac{1}{2\pi}\int_{-\pi/8}^{\pi/8} e^{-j3\omega} \cdot e^{j3\omega}\, d\omega = \frac{1}{2\pi} \times \left[\frac{\pi}{8} - \left(-\frac{\pi}{8}\right)\right]$$

$$= \frac{1}{2\pi} \times \frac{2\pi}{8} = \frac{1}{8}$$

So,

$$h_d[n] = \begin{cases} \dfrac{\sin\left[\frac{\pi}{8}(n-3)\right]}{\pi(n-3)}, & n \neq 3 \\ \dfrac{1}{8}, & n = 3 \end{cases}$$

In order to design linear-phase FIR filter,

$$h[n] = h[M-1-n] = h[10-n]$$

desired filter coefficients are calculate for,

$\cancel{n = 0, 1, 2, \ldots, \frac{M-1}{2} = 0, 1, 2, \ldots, 5}$

$$-\left(\frac{M-1}{2}\right) \le n \le \left(\frac{M+1}{2}\right)$$

$$-5 \le n \le 5$$

Then,

$h_d[0] = 0.098$

$h_d[1] = 0.1125 = h_d[-1]$

$h_d[2] = 0.1218 = h_d[-2]$

$h_d[3] = 1/8 = 0.125 = h_d[-3]$

$h_d[4] = 0.1218 = h_d[-4]$

$h_d[5] = 0.1125 = h_d[-5]$

Step 3 :

ⓐ For Rectangular window

$$w_R[n] = \begin{cases} 1, & -\frac{(N-1)}{2} \le n \le \frac{M-1}{2} \\ 0, & \text{otherwise} \end{cases}$$ (caus

$$= \begin{cases} 1, & -5 \le n \le 5 \\ 0, & \text{otherwise} \end{cases}$$

~~As, h[n] = h[M-1-n] = h[10-n]~~

~~⇒ $h_d[n] = h_d[10-n]$~~

∴ ~~h[0] = h[10]~~

Then, the actual filter coefficients are calculated f ~~n = 0, 1, 2, 3, 4, 5~~

$w_R[0] = 1$

$w_R[1] = 1 = w_R[-1]$

$w_R[2] = 1 = w_R[-2]$

$w_R[3] = 1 = w_R[-3]$

$w_R[4] = 1 = w_R[-4]$

$w_R[5] = 1 = w_R[-5]$

Now, The impulse response of FIR filter is,

$$h[n] = h_d[n] \cdot w_R[n]$$

But, $h[n] = h[M-1-n] = h[11-1-n] = h[10-n]$

$\therefore h[n] = h[10-n]$

So, The impulse response of FIR filter is,

$$h[n] = h_d[n] \cdot w_R[n]$$

So,

For $n=0$, $h[0] = h[10] = h_d[0] \cdot w_R[0] = 0.098$

$n=1$, $h[1] = h[9] = h_d[1] \cdot w_R[1] = 0.1125$

$n=2$, $h[2] = h[8] = h_d[2] \cdot w_R[2] = 0.1218$

$n=3$; $h[3] = h[7] = h_d[3] \cdot w_R[3] = 0.125$

$n=4$, $h[4] = h[6] = h_d[4] \cdot w_R[4] = 0.1218$

$n=5$, $h[5] = h[5] = h_d[5] \cdot w_R[5] = 0.1125$

KAISER WINDOW NUMERICALS

Example 1: The desired specification of an FIR LPF is given by $\alpha_p = 1dB$ at $\omega_p = 20 rad/sec$ and $\alpha_s = 39dB$ at $\omega_s = 40 rad/sec$. Design a FIR LPF using KAISER WINDOW technique. The sampling frequency is $\omega_{SF} = 100 rad/sec$.

→ Given that

Pass-band Attenuation, $\alpha_p = 1dB$

stopband Attenuation, $\alpha_s = 39dB$

Sampling frequency, $\omega_{SF} = 100 rad/sec$

In Hz,

$$\text{Sampling frequency, } F_{SF} = \frac{\omega_{SF}}{2\pi} = \frac{100}{2\pi} \text{ Hz}$$

passband edge frequency, $\omega_p' = 20 rad/sec$

After normalization,

$$\omega_p = \frac{\omega_p'}{F_{SF}} = \frac{20}{100/2\pi} = \frac{4\pi}{10} = 0.4\pi \text{ rad}$$

stopband edge frequency, $\omega_s' = 40 rad/sec$

After normalization,

$$\omega_s = \frac{\omega_s'}{F_{SF}} = \frac{40}{100/2\pi} = 0.8\pi \text{ rad.}$$

Design steps

Step 1: Determine ripple parameters in respective band, as:

$$\delta_s = 10^{-0.05\alpha_s} = 10^{-0.05\times 39} = 0.01122$$

$$\delta_p = \frac{10^{0.05\alpha_p}-1}{10^{0.05\alpha_s}-1} = \frac{10^{0.05\times 1}-1}{10^{0.05\times 1}+1} = 0.0575$$

From calculation, $\delta_s < \delta_p$

∴ We choose, $\delta = \delta_s = 0.01122$

Step 2: Calculate the Attenuation in dB as,

$$\alpha = -20\log\delta = -20\log(0.01122) = 39 \text{ dB}$$

allows adjustment of the Compromise between the overshoot reduction & transition region (breadth) width

Step 3: Determining the Kaiser Parameter (β) as:

$$\beta = 0.5842(\alpha-21)^{0.4} + 0.07886(\alpha-21)$$

$$= 0.5842(39-21)^{0.4} + 0.07886(39-21)$$

$$= 3.276$$

Step 4: Order of the filter, M is calculated as

$$M = \frac{\alpha-8}{2.285\Delta\omega} = \frac{39-8}{2.285(\omega_s-\omega_p)}$$

$$= \frac{31}{2.285(0.8\pi-0.4\pi)}$$

$$= \cancel{8.1288}\ 10.79$$

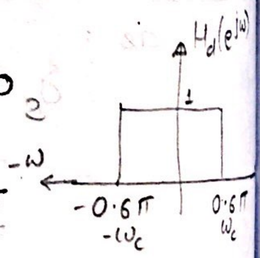
$\therefore M \approxeq 11 \Rightarrow$ symmetric FIR filter with odd length

$\therefore$ Range of filter: $-\frac{M-1}{2} \le n \le \frac{M-1}{2}$

$$\Rightarrow -5 \le n \le 5$$

Step 5: Low-Pass filter is to be designed whose desired impulse response is given as (derived earlier) [ideal LPF]

$$h_d[n] = \begin{cases} 2F_c\frac{\sin(\omega_c n)}{\omega_c n} = \frac{\sin(\omega_c n)}{\pi n}, & -5 \le n \le 5\ (n \ne 0) \\ 2F_c, & n = 0 \end{cases}$$



Where, $\omega_c = \frac{\omega_s+\omega_p}{2} = \frac{0.8\pi+0.4\pi}{2} = 0.6\pi$

$$\Rightarrow F_c = \frac{0.6\cancel{\pi}}{2\cancel{\pi}} = 0.3$$

So, $$h_d[n] = \begin{cases} \frac{\sin(0.6\pi n)}{\pi n}, & -5 \le n \le 5\ (n \ne 0) \\ 2F_c = 2\times0.3 = 0.6, & n = 0 \end{cases}$$

The desired impulse response coefficients are,

$h_d[0] = 0.6$, $h_d[1] = h_d[-1] = 0.303$, $h_d[2] = h_d[-2] = -0.03$

$h_d[3] = h_d[-3] = -0.06$, $h_d[4] = h_d[-4] = 0.076$

$h_d[5] = h_d[-5] = 0$

The kaiser window function is define as,

$$W_k[n] = \begin{cases} \dfrac{I_0\left[\beta\sqrt{1-\left(\dfrac{2n}{M-1}\right)^2}\right]}{I_0(\beta)} & , -5 \le n \le 5 \\ 0 & , \text{otherwise} \end{cases}$$

$$= \begin{cases} \dfrac{I_0\left[3.276\sqrt{1-\left(\dfrac{2n}{10}\right)^2}\right]}{I_0(3.276)} & , -5 \le n \le 5 \\ 0 & , \text{otherwise} \end{cases}$$

Where,

$$I_0(\beta) = 1 + \frac{0.25\beta^2}{(1!)^2} + \frac{(0.25\beta^2)^2}{(2!)^2} + \frac{(0.25\beta^2)^3}{(3!)^2} + \cdots$$

(We approximate to 3rd decimal places only)

$$A = \cancel{3.276\left(1-\left(\frac{2n}{10}\right)\right)} \left[3.276\sqrt{1-\left(\frac{2n}{10}\right)^2}\right]$$

On varying $n : -5 \le n \le 5$

$$A = \{\underset{-5}{0}, \underset{-4}{1.9656}, \underset{-3}{2.6208}, \underset{-2}{3.0025}, \underset{-1}{3.2098}, \underset{0}{3.276}, \underset{1}{3.2098}, \underset{2}{3.0025}, \underset{3}{2.6208}, \underset{4}{1.9656}, \underset{5}{0}\}$$

Now,

$$W_k[-5] = W_k[5] = \frac{I_0(0)}{I_0(3.276)} = \frac{1}{6.019} = 0.166$$

$$W_k[-4] = W_k[4] = \frac{I_0(1.9656)}{6.019} = \frac{2.224}{6.019} = 0.368$$

$$W_k[-3] = W_k[3] = \frac{I_0(2.6208)}{6.019} = \frac{3.595}{6.019} = 0.59$$

$$W_k[-2] = W_k[2] = \frac{I_0(3.0025)}{6.019} = \frac{4.842}{6.019} = 0.796$$

$$w_k[-1] = w_k[1] = \frac{I_0(3.2098)}{6.019} = \frac{5.709}{6.019} = 0.947$$

$$w_k[0] = \frac{I_0(3.276)}{6.019} = \frac{6.019}{6.019} = 1$$

Now,

The Filter Coefficients obtained due to Kaiser windo

$$h[n] = h_d[n] \times w_k[n]$$

For, $-5 \le n \le 5$, & using the Condition, $h[n] = h[M-1-n]$ $= h[10-n]$

$$h[0] = h_d[0] \times w_k[0] = h[10] = 0.6 \times 1 = 0.6$$

$$h[1] = h_d[1] \times w_k[1] = h[9] = 0.303 \times 0.947 = 0.287$$

$$h[2] = h_d[2] \times w_k[2] = h[8] = -0.094 \times 0.796 = -0.075$$

$$h[3] = h_d[3] \times w_k[3] = h[7] = -0.062 \times 0.59 = -0.037$$

$$h[4] = h_d[4] \times w_k[4] = h[6] = 0.076 \times 0.362 = 0.028$$

$$h[5] = h_d[5] \times w_k[5] = h[5] = 0 \times 0.166 = 0$$

The FIR filter T

...mple 2 : Design an FIR Linear phase Filter using KAISER window to meet the following specifications:

$$0.98 \le |H(e^{j\omega})| \le 1.02 \quad \text{for } 0 \ge \omega \ge 0.9\pi$$

$$|H(e^{j\omega})| \le 0.01 \quad \text{for } 0.14\pi \le \omega \le \pi$$

According to the given specification of frequency, it is HPF.

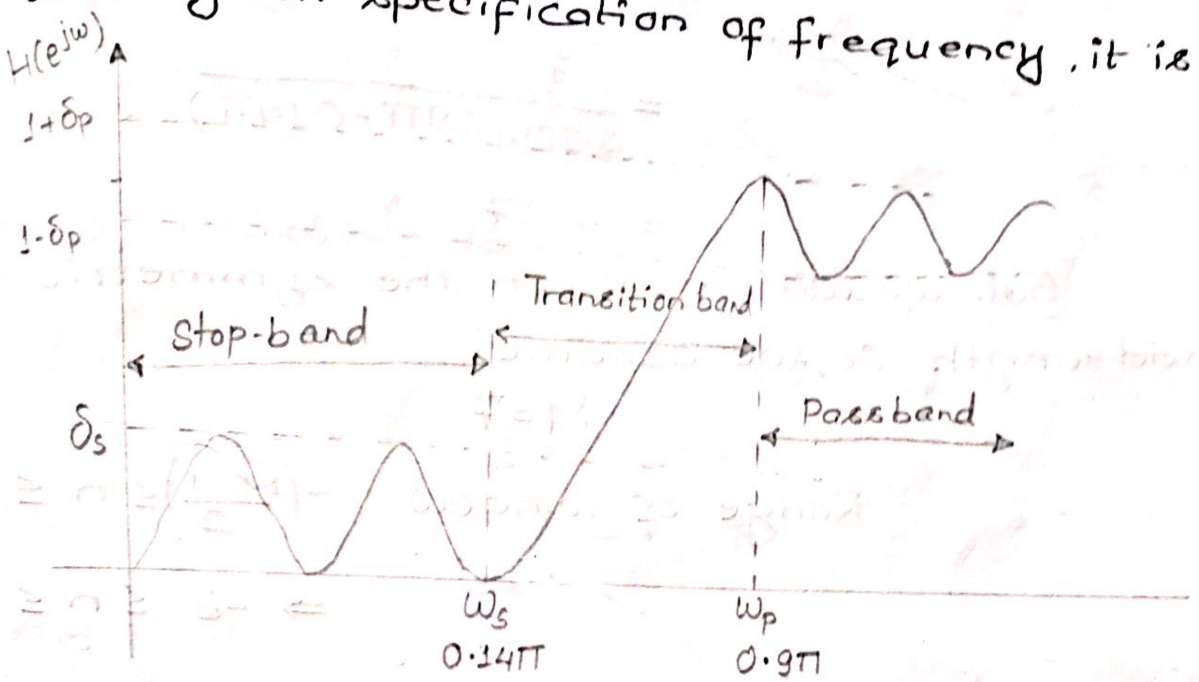


Fig: HPF magnitude spectrum

Then,

Pass-band frequency, $\omega_p = 0.9\pi$

Stopband frequency, $\omega_s = 0.14\pi$

Comparing the given eqn with

$$1 - \delta_p \le |H(e^{j\omega})| \le 1 + \delta_p$$

$$\Rightarrow 1 - 0.02 \le |H(e^{j\omega})| \le 1 + 0.02$$

$$\delta_p = 0.02$$

and, $|H(e^{j\omega})| \le \delta_s$

$\Rightarrow \delta_s = 0.01$

Design steps

Step 1 : Chose optimum value of ripple so

$\delta_s < \delta_p$

$\Rightarrow \delta = \delta_s = 0.01$

Step 2 : Calculate the attenuation in dB,

$$\alpha = -20\log(\delta)$$
$$= -20\log(0.01)$$
$$= 40\,dB$$

Step 3 : Determine the kaiser parameter (β)

$$\beta = 0.5842(\alpha-21)^{0.4} + 0.07886(\alpha-21)$$
$$= 3.39$$

Step 4 : Order of the filter (M) is,

$$M = \frac{\alpha-8}{2.285\Delta\omega} = \frac{40-8}{2.285(\omega_p-\omega_s)}$$

$$= \frac{32}{2.285(0.9\pi - 0.14\pi)}$$

$$= 5.87$$

But, we wish to design the symmetric FIR-filter odd length. So, we assume,

$$M = 7$$

$\therefore$ Range of window : $-\left(\frac{M-1}{2}\right) \le n \le \frac{M-1}{2}$

$\Rightarrow -3 \le n \le 3$

Step 5 : The kaiser window function is defined as,

$$w_k[n] = \begin{cases} \dfrac{I_0\left\{\beta\sqrt{1-\left(\frac{2n}{M-1}\right)^2}\right\}}{I_0(\beta)} & , -3 \le n \le 3 \\ 0 & , \text{otherwise} \end{cases}$$

$$= \begin{cases} \dfrac{I_0\left\{3.39\sqrt{1-\left(\frac{2n}{6}\right)^2}\right\}}{I_0(3.39)} & , -3 \le n \le 3 \\ 0 & , \text{otherwise} \end{cases}$$

Where,

$$I_0(\beta) = 1 + \frac{0.25\beta^2}{(1!)^2} + \frac{(0.25\beta^2)^2}{(2!)^2} + \frac{(0.25\beta^2)^3}{(3!)^2}$$

(Approximate to 3rd decimal place only)

let,

$$A = \left\{3.39\sqrt{1-\left(\frac{2n}{6}\right)^2}\right\}$$

On varying $-3 \le n \le 3$

$$A(-3) = A(3) = 0$$
$$A(-2) = A(2) = 2.526$$
$$A(-1) = A(1) = 3.196$$
$$A(0) = 3.39$$

Now,

$$\omega_K[-3] = \omega_K[3] = \frac{I_0(0)}{I_0(3.39)} = \frac{1}{6.59} = 0.152$$

$$\omega_K[-2] = \omega_K[2] = \frac{I_0(2.526)}{I_0(3.39)} = \frac{3.344}{6.59} = 0.507$$

$$\omega_K[-1] = \omega_K[1] = \frac{I_0(3.196)}{I_0(3.39)} = \frac{5.64}{6.59} = 0.856$$

$$\omega_K[0] = \frac{I_0(3.39)}{I_0(3.39)} = \frac{6.59}{6.59} = 1$$

Step 6 : HPF is to be designed whose desired impulse response is, (Ideal HPF) ⇒ derived earlier

????

$\tau = 3$

$$h_d[n] = \begin{cases} -2F_c \dfrac{\sin(\omega_c n)}{\omega_c n} & , -3 \le n \le 3 \ (n \ne 0 \\ 1 - 2F_c & , n = 0 \end{cases}$$

$$= \begin{cases} -\dfrac{\sin(\omega_c n)}{\pi n} & , -3 \le n \le 3 \ (n \ne 0) \\ 1 - 2F_c & , n = 0 \end{cases}$$

RC

Where, $\omega_c = \frac{\omega_p + \omega_s}{2} = \frac{0.9\pi + 0.14\pi}{2} = 0.52\pi$

$\Rightarrow F_c = \frac{\omega_c}{2\pi} = \frac{0.52\pi}{2\pi} = 0.26$

Now,

$$h_d[n] = \begin{cases} \frac{\sin(0.52\pi n)}{\pi n}, & -3 \le n \le 3 \ (n \neq 0 \\ 1 - 2F_c = 1 - 2 \times 0.26 = 0.48, & n = 0 \end{cases}$$

Now,

$h_d[-3] = -0.104 = h_d[3]$

$h_d[-2] = -0.02 = h_d[2]$

$h_d[-1] = 0.317 = h_d[1]$

$h_d[0] = 0.48$

Now, The Filter coefficients obtained due to kaiser window are :

$h[n] = h_d[n] \times \omega_k[n]$ For $-3 \le n \le 3$ & using the condition $h[n] = h[M-1-n]$ $= h[6-n]$

Where,

$h[0] = h_d[0] \times W_k[0] = 0.48 \times 1 = 0.48 = h[6]$

$h[1] = h_d[1] \times W_k[1] = 0.317 \times 0.856 = 0.271 = h[5]$

$h[2] = h_d[2] \times W_k[2] = -0.02 \times 0.507 = -0.0101 = h[4]$

$h[3] = h_d[3] \times W_k[3] = -0.104 \times 0.152 = -0.016$

The FIR-Filter

$-0.202\cos 2\omega - 0.032\cos 3\omega$

: Design the FIR filter using suitable window the following specifications :

$$0.899 \le |H(e^{j\omega})| \le 1 \quad ; \text{for } |\omega| \le 0.2\pi$$

$$|H(e^{j\omega})| \le 0.01 \; ; \text{for } 0.4\pi \le |\omega| \; \pi$$

ording to the frequency specification it is a LPF where,

pass-band frequency, $\omega_p = 0.2\pi$

stopband frequency, $\omega_s = 0.4\pi$

passband ripple, $\delta_p = 0.899$

stop-band ripple, $\delta_s = 0.01$.

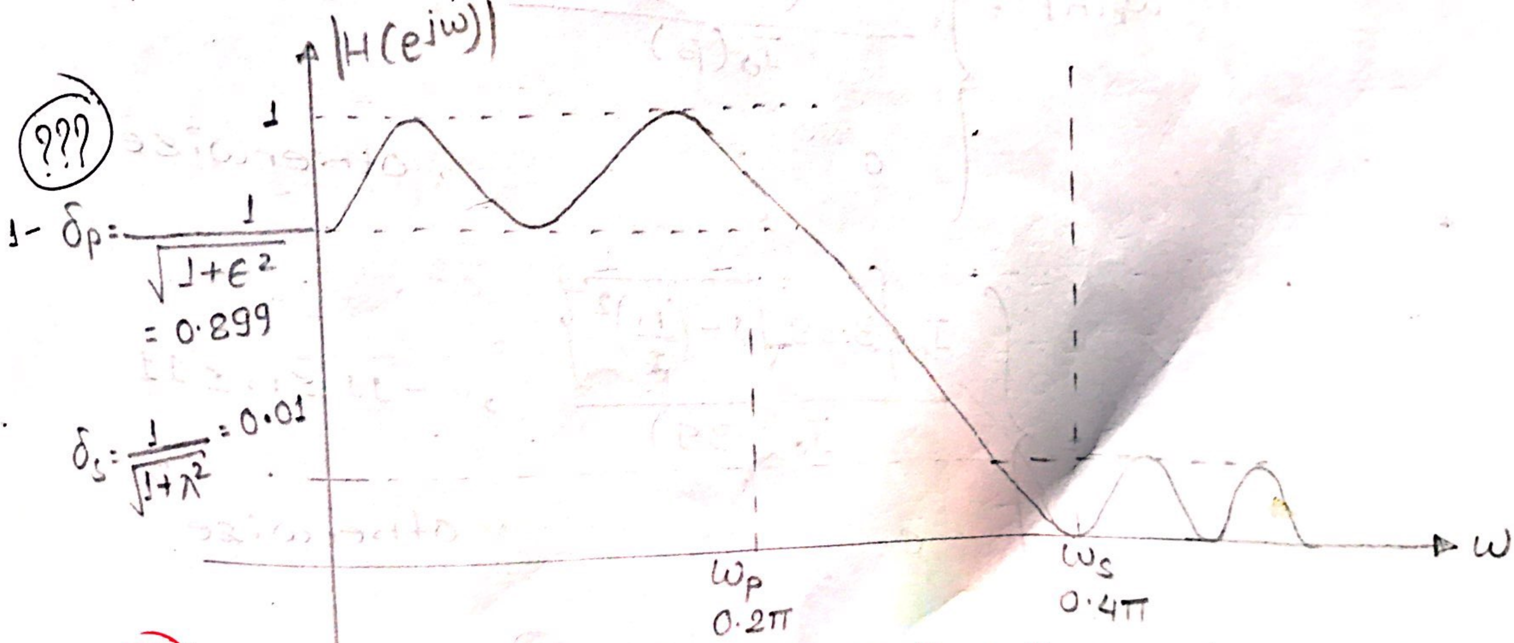


Fig : Normalized LPF Magnitude Spectrum

87

Design steps

Step 1: chose the optimum value of ripple, so,

$$\delta_s < \delta_p$$???

$$\Rightarrow \delta = \delta_s = 0.01$$

Step 2: Calculate the attenuation in dB.

$$\alpha = -20\log(\delta)$$
$$= -20\log(0.01)$$
$$= 40\,dB$$

Step 3: Calculate the KAISER PARAMETER

$$\beta = 0.5842(\alpha-21)^{0.4} + 0.7886(\alpha-21)$$
$$= 0.5842(40-21)^{0.4} + 0.7886(40-21)$$
$$= 3.39$$

Step 4: order of the filter (M) is,

$$M = \frac{\alpha - 8}{2.285\,\Delta\omega} = \frac{40-8}{2.285(\omega_s - \omega_p)}$$

$$= \frac{32}{2.285(0.4\pi - 0.2\pi)} = 22.28$$

let us assume we design a symmetric FIR filter with odd-length. $\therefore M = 23$

$\therefore$ Range of window : $-\left(\frac{M-1}{2}\right) \le n \le \left(\frac{M-1}{2}\right)$

$$\Rightarrow -11 \le n \le 11$$

Step 5: The kaiser window function is defined as:

$$w_k[n] = \begin{cases} \dfrac{I_0\left\{\beta\sqrt{1-\left(\frac{2n}{M-1}\right)^2}\right\}}{I_0(\beta)} & , -11 \le n \le 11 \\ 0 & , \text{otherwise} \end{cases}$$

$$= \begin{cases} \dfrac{I_0\left\{3.39\sqrt{1-\left(\frac{n}{11}\right)^2}\right\}}{I_0(3.39)} & , -11 \le n \le 11 \\ 0 & , \text{otherwise} \end{cases}$$

Let, $A = \left\{ 3.39 \sqrt{1 - \left(\frac{n}{11}\right)^2} \right\}$

$A(-11) = A(11) = 0$

$A(-10) = A(10) = 1.412$

$A(-9) = A(9) = 1.95$

$A(-8) = A(8) = 2.326$

$A(-7) = A(7) = 2.615$

$A(-6) = A(6) = 2.84$

$A(-5) = A(5) = 3.019$

$A(-4) = A(4) = 3.157$

$A(-3) = A(3) = 3.26$

$A(-2) = A(2) = 3.33$

$A(-1) = A(1) = 3.376$

$A(0) = 3.39$

Now, The window filter coefficients,

$$w_k[-11] = w_k[11] = \frac{I_0(0)}{I_0(3.39)} = \frac{1}{6.59} = 0.1517$$

$w[-10] = w[10] = 0.237$

$w[-9] = w[9] = 0.333$

$w[-8] = w[8] = 0.437$

$w[-7] = w[7] = 0.543$

$w[-6] = w[6] = 0.646$

$w[-5] = w[5] = 0.744$

$w[-4] = w[4] = 0.83$

$w[-3] = w[3] = 0.902$

$w[-2] = w[2] = 0.953$

$w[-1] = w[1] = 0.989$

$w[0] = w[0] = 1$

Step 6. Ideal LPF whose designed impulse response is

????

$$h_d[n] = \begin{cases} \dfrac{\sin(\omega_c n)}{\pi n}, & -11 \le n \le 11 \ (n \ne 0) \\ 2F_c, & n = 0 \end{cases}$$

Where, $\omega_c = \dfrac{\omega_p + \omega_s}{2} = \dfrac{0.2\pi + 0.4\pi}{2} = 0.3\pi$

$$\Rightarrow F_c = \frac{\omega_c}{2\pi} = \frac{0.3\cancel{\pi}}{2\cancel{\pi}} = 0.15$$

$$\therefore h_d[n] = \begin{cases} \dfrac{\sin(0.3\pi n)}{\pi n}, & -11 \le n \le 11 \\ 2F_c = 2\times 0.15 = 0.3, & n = 0 \end{cases}$$

So, The desired impulse response of LPF is,

$h_d[-11] = h_d[11] = -0.023$

$h_d[-10] = h_d[10] = 0$

$h_d[-9] = h_d[9] = 0.0286$

$h_d[-8] = h_d[8] = 0.037$

$h_d[-7] = h_d[7] = 0.014$

$h_d[-6] = h_d[6] = -0.031$

$h_d[-5] = h_d[5] = -0.063$

$h_d[-4] = h_d[4] = -0.046$

$h_d[-3] = h_d[3] = 0.032$

$h_d[-2] = h_d[2] = 0.151$

$h_d[-1] = h_d[1] = 0.257$

$h_d[0] = 0.3$

Now,
The Filter Coefficients obtained due to kaiser window is,

$$h[n] = h_d[n] \times w[n]$$

for $-11 \leq n \leq 11$ & using the condition,

S.No.	n	$h_d[n]$	$w[n]$	$h[n] = h_d[n] \times w[n]$	$h[n] = h[M-1-n] = h[22-n]$
1	+11	-0.023	0.1517	-3.489×10^{-3}	$h[22]$
2	+10	0	0.237	0	$h[21]$
3	+9	0.0286	0.333	9.52×10^{-3}	$h[20]$
4	+8	0.037	0.437	0.016	$h[19]$
5	+7	0.014	0.543	7.602×10^{-3}	$h[18]$
6	+6	-0.031	0.646	-0.02	$h[17]$
7	+5	-0.063	0.744	-0.0468	$h[16]$
8	+4	-0.046	0.83	-0.0382	$h[15]$
9	+3	0.032	0.902	0.0288	$h[14]$
10	+2	0.151	0.953	0.144	$h[13]$
11	+1	0.257	0.989	0.254	$h[12]$
12	0	0.3	1	0.3	$h[11]$

FIR FILTER DESIGN USING FREQUENCY SAMPLING TECHNIQUE

xxx

→ The frequency sampling technique is another method for designing FIR filters. In this technique, a set of samples is determined from a desired frequency response and these samples are identified as discrete Fourier transform (DFT) coefficients. The signal can be determined by taking Inverse Discrete Fourier Transform (IDFT) of the samples.

The impulse response $h[n]$ (Filter Coefficients) of an FIR filter is obtained by taking IDFT of the frequency response $H(K)$, i.e;

$$h_d[n] = \frac{1}{N} \sum_{K=0}^{M-1} H_d(e^{j\omega}) e^{j\frac{2\pi}{N}nK} \quad , n = 0,1,2,3, \ldots, (M-1)$$

Where,

$$H_d(e^{j\omega}) = \sum_{n=0}^{M-1} h_d[n] e^{-j\frac{2\pi}{N}nK} \quad , K = 0,1,2,3, \ldots, (M-1)$$

The set of samples points can be obtained by sampling the desired frequency response $H_d(e^{j\omega})$ at N-points, equally spaced around the unit circle.

$$H_d(e^{j\omega}) = H(z) \Big|_{z = e^{j\left(\frac{2\pi}{M}\right)K}}$$

The Transfer function of FIR filter is given by,

$$H_d(z) = \sum_{n=0}^{M-1} h_d[n] z^{-n}$$

The frequency samples are choosen to be,

$$\omega_k = \frac{2\pi}{M} K \quad , K = 0,1,2, \ldots, M-1$$

Therefore, the sampled frequency response is given by,

$$H(K) = H_d(e^{j\omega}) \Big|_{\omega = \omega_K} \quad , K = 0,1,2,3, \ldots, (M-1)$$

$$H(k) = H_d(e^{j\omega}) \Big|_{\omega = \frac{2\pi}{M}K} \quad , K = 0,1,2,3, \ldots (M-1)$$

Design

While designing using frequency sampling technique, we the symmetry property of the sampled frequency respon. The sampled frequency response $H_d(e^{j\omega})$ as :

$$H(k) = H_d(e^{j\omega})\Big|_{\omega=\frac{2\pi}{M}k} \quad , k = 0,1,2,\ldots\ldots, M-1$$

Here, The spacing between two samples is $\frac{2\pi}{M}$

But, In magnitude & phasor form

$$H(k) = |H(k)|\, e^{j\theta(\omega)}$$

Where, For linear-phase FIR filter,

$$\theta(k) = -\alpha\omega\Big|_{\omega=\frac{2\pi}{M}k}$$

i.e, $$\theta(k) = -\left(\frac{M-1}{2}\right)\frac{2\pi}{M}k$$

$$= -\left(\frac{M-1}{M}\right)\pi k \quad , k = 0,1,2,3,\ldots\ldots,(N-1)$$

Now, The FIR filter Coefficients $h[n]$ of a FIR filter must be real. Therefore, all the Complex term appear in complex Conjugate pairs.

For real $H(k)$ and odd M, the frequency response become

$$H(M-k) = H^*(k) \quad , k = 0,1,2,3,\ldots\ldots,(M-1)$$

For real $H(k)$ and even M, the frequency response becom

$$H(M-k) = H^*(k) \quad , k = 0,1,2,3,\ldots\ldots,(M-1)$$

and, $$H\left(\frac{M}{2}\right) = 0$$

Since, phase is an odd function, eqn 1 becomes,

$$\theta(k) = -\theta(M-k)$$

$$= +\left(\frac{M-1}{M}\right)\pi\cdot(M-k) \quad , k = 0,1,2,\ldots\ldots(M-1)$$

Here,

The R.H.S of equation (7) is simply the DFT of $H(k)$.

Therefore, eqn (7) can be rewritten as,

$$H\left(e^{j\frac{2\pi}{M}k}\right) = H(k) = H_d\left(e^{j\frac{2\pi}{M}k}\right)$$

only at $\omega = \frac{2\pi}{M}k$, the value of $H(k)$ & $H_d(k)$ coincides.

Remez Exchange Algorithm

(i) Initialize and estimate for the extreme frequencies $\omega_0, \omega_1, \ldots\ldots, \omega_{L+1}$ by selecting $(L+2)$ equally spaced frequencies at both passband and stop-band.

(ii) Find $P(\omega_k)$ and δ such that,

$$W_q(\omega_k) = \left[D_q(\omega_k) - P(\omega_k)\right] = (-1)^k \delta, \quad k = 0, 1, 2, \ldots\ldots, (L+1)$$

An alternate and efficient approach to compute δ is,

$$\delta = \frac{a_0 D_q(\omega_0) + a_1 D_q(\omega_1) + \cdots\cdots + a_{L+1} D_q(\omega_{L+1})}{\dfrac{a_0}{W_q(\omega_0)} + \dfrac{a_1}{W_q(\omega_1)} + \dfrac{a_2}{W_q(\omega_2)} - \cdots\cdots + \dfrac{(-1)^{L+1} a_{L+1}}{W_q(\omega_{L+1})}}$$

Where,

$$a_k = \prod_{\substack{i=0 \\ i \neq k}}^{L+1} \frac{1}{(\cos\omega_k - \cos\omega_i)}$$

(iii) $P(\omega)$ can be computed by using Lagranges interpolation technique.

$$p(\omega) = \begin{cases} C_k & , \omega = \omega_0, \omega_1, \ldots, \omega_L \\[2ex] \dfrac{\displaystyle\sum_{k=0}^{L} \left(\frac{\beta_k}{\cos\omega - \cos\omega_k}\right) C_k}{\displaystyle\sum_{k=0}^{L} \frac{\beta_k}{(\cos\omega - \cos\omega_k)}} & , \omega \neq \omega_0, \omega_1, \ldots, \omega_L \end{cases}$$

Where,

$$C_k = D_q(\omega_k) - \frac{(-1)^k \delta}{W_q(\omega_k)}$$

$$\beta_k = \prod_{\substack{i=0 \\ i \neq k}}^{L} \frac{1}{(\cos\omega_k - \cos\omega_i)} = a_k(\cos\omega_k - \cos\omega_{L+1})$$

For $k = 0, 1, 2, \ldots\ldots, (L+1)$

(iv) Evaluate $|E(\omega)|$ for various frequencies. If $|E(\omega)| \leq |\delta|$ for all frequencies, the optimal solution has been found.

(v) If $|E(\omega)| > |\delta|$ for some frequencies, a new set of extremes must be chosen as the peaks of $E(\omega)$. Allow δ to grow and converge to its upper limit. If the number of extremes are more than (L+2) in $E(\omega)$, keep the location of the (L+2) largest values of the extremes of $|E(\omega)|$ and return to the step(ii). The optimum solution in terms of

$$H(\omega) = p(\omega) \cdot Q(\omega) \text{ can be found}$$

where, $p(\omega)$ & $Q(\omega)$ are defined in Table ...... before.

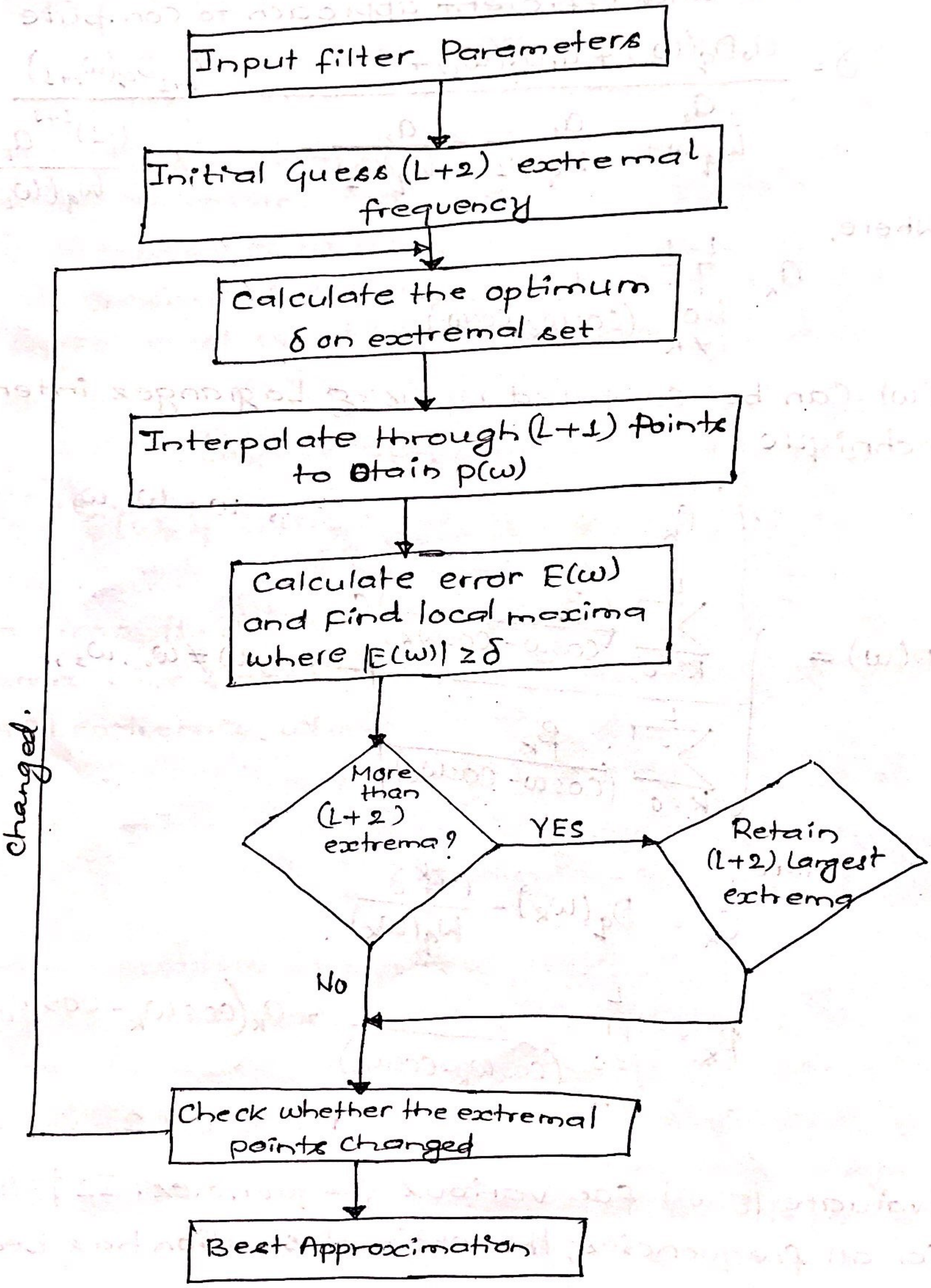


Fig: Flowchart of Remez Algorithm.